Classification of superconductors

Superconductors are basically materials that conduct electricity with zero resistance and expel magnetic fields when cooled below a specific critical temperature. This cooling process is done with the help of other materials, and their costs vary greatly. That is why their classification into high and low temperature ones with predefined threshold 77K is so important. This threshold was chosen because of the boiling point of liquid nitrogen which high temperature superconductors can use for cooling. But low temperature superconductors need to use liquid helium which is around 50x more expensive. High temperature superconductors are also rare, that is why their discovery is important, but their properties need to be verified in a laboratory experiment which costs time and resources.

Hypothesis: Material properties alone (atomic mass, valence, thermal conductivity, etc.) are enough to reliably classify a superconductor as high/low temperature one without measuring its critical temperature directly.

To test this hypothesis, we used a dataset that consists of 21,263 superconducting materials. We trained four different machine learning models that learn patterns from the material properties to predict whether a material is a high/low temperature superconductor. Four classifier models were trained and compared: Logistic Regression, Support Vector Machine, Random Forest and Decision Tree. Each model was trained in goal to find as many high temperature superconductors as possible, while keeping the number of false candidates low enough to remain practical for laboratory validation.

Model	HTc Superconductors found	Missed	Unnecessary tests
LR-B	767 / 791 (97%)	24	774
Random Forest	755 / 791 (95%)	36	235
Decision Tree	717 / 791 (91%)	74	323

Random Forest performed the best for this classification task with most stable predictions missing only 36 high temperature superconductors. LR-B was even better in this way, missing only 24, but with a bad tradeoff giving us 539 more false ones which results in unnecessary tests. This would waste a significant number of resources and time in lab validation experiments when testing if material is truly a high temperature superconductor. The results confirm our hypothesis: material properties alone are sufficient to reliably identify high-temperature superconductors, and Random Forest provides the best practical balance between discovering new high temperature materials and efficiency.